Brownian sticky clocks: construction, holding time and causal recovery in the observed filtration

Project Stardust
A companion to a frozen verified library

October 29, 2026

Abstract

We reconstruct the complete construction of a Brownian sticky time change from smooth convex approximations to absolute value. A compact probability estimate and a summable subsequence give a measurable continuous increasing mass, whose Stieltjes measure is supported on Brownian zeros. Its inverse clock has an exact physical holding-time identity. The clock is recoverable from the observed path up to the current time, so both the process and its compensated square are martingales in explicitly specified observed filtrations. At every positive parameter and deterministic horizon, expected holding is positive and the second moment is strictly smaller than elapsed time. An independent Brownian source on the complementary clock yields two genuine Brownian coordinate processes whose covariance equals expected holding. Their finite-increment laws are proved by stopped Fourier products, rather than inferred from Gaussian one-time marginals. These classical mechanisms are presented with full proofs and exact formal-source correspondences. The construction does not use the separately unfinished closed-generator or stochastic-flow identification.

1 The construction and its precise conclusions

Let $B = (B_s)_{s\geq 0}$ be standard real Brownian motion, $B_0 = 0$, with a measurable realization in $C([0,\infty),\mathbb{R})$. We use its raw natural filtration, augmented by ambient measurable null events:

$$\mathcal{F}_s = \sigma(B_u : 0 \leq u \leq s) \vee \mathcal{N}.$$

Here $\mathcal{N}$ consists of the null events in the ambient probability space; this convention suffices for all modifications below. No right-continuous enlargement of the filtration is implicit. The Brownian stochastic integral is its ordinary predictable $L^2(ds\,d\mathbb{P})$ isometry, with conditional expectations at deterministic horizons. These standard Brownian foundations are also constructed in the frozen source library.

Theorem 1.1. *There is a measurable continuous nonnegative increasing process L, starting from zero, with the following properties. For each fixed s,*

$$L_s = |B_s| - \int_0^s \operatorname{sgn}(B_u)\, dB_u \quad \text{in } L^2. \tag{1}$$

Its Stieltjes measure dL is concentrated on $\{s : B_s = 0\}$. For every $\kappa \geq 0$, set

$$A_\kappa(s) = s + \kappa L_s, \qquad C_\kappa = A_\kappa^{-1}, \qquad X_t^\kappa = B_{C_\kappa(t)}. \tag{2}$$

The clock and process have measurable continuous path realizations. For almost every sample, simultaneously for all $\kappa, t \geq 0$,

$$H_\kappa(t) := \int_0^t \mathbf{1}_{\{X_r^\kappa = 0\}}\, dr = \kappa L_{C_\kappa(t)} = t - C_\kappa(t). \tag{3}$$

For each fixed κ, write

$$\mathcal{G}_t^0 = \sigma(X_r^\kappa : 0 \leq r \leq t), \qquad \mathcal{G}_t = \mathcal{G}_t^0 \vee \mathcal{N}.$$

The operational clock $C_\kappa(t)$ is $\mathcal{G}_t$-measurable. The process X^κ is a martingale in both $\mathcal{G}^0$ and $\mathcal{G}$, and $(X_t^\kappa)^2 - C_\kappa(t)$ is a martingale in $\mathcal{G}$. For $\kappa, t > 0$,

$$\mathbb{E}H_\kappa(t) > 0, \qquad \mathbb{E}(X_t^\kappa)^2 + \mathbb{E}H_\kappa(t) = t, \qquad \mathbb{E}(X_t^\kappa)^2 < t. \tag{4}$$

In particular, $\mathbb{P}(H_\kappa(t) > 0) > 0$.

The last assertion is positive probability, rather than an assertion that holding is positive on every sample or that a terminal atom has been computed at each time. At $\kappa = 0$, $C_0(t) = t$ and $X^0 = B$. All clocks in the theorem use the same Brownian sample. Independence of the clock and the Brownian motion is neither true nor needed.

The time-change construction is classical. The appendix of Ayala–Carinci–Redig [1], equations (112)–(114), explicitly uses Brownian time plus a local-time term and its inverse to construct two-sided sticky Brownian motion. Bass [2], Section 2, equations (2.3)–(2.4), places inverse additive clocks in the classical speed-measure construction. Our contribution is an exposition of the verified construction, its versions and observed-filtration statements; we make no independent priority claim for these mechanisms.

2 Smooth curvature masses and their fixed-time limit

For $e > 0$ let

$$r_e(x) = \sqrt{x^2 + e^2}, \qquad a_e(x) = \frac{x}{\sqrt{x^2 + e^2}}, \qquad b_e(x) = \frac{e^2}{(x^2 + e^2)^{3/2}}.$$

Direct differentiation gives $r_e' = a_e$, $r_e'' = b_e$. Moreover,

$$0 \leq r_e(x) - |x| \leq e, \quad |a_e(x)| \leq 1, \quad 0 \leq b_e(x) \leq e^{-1}. \tag{5}$$

Indeed, $|x| \leq r_e(x) \leq |x| + e$ follows by squaring nonnegative quantities. The slope bound follows from $|x| \leq r_e$; the curvature bound follows from $r_e \geq e$. Fix the positive sequence $e_n \downarrow 0$ used in the source, and put

$$L_t^n = \frac{1}{2}\int_0^t b_{e_n}(B_u)\, du, \qquad M_t^n = r_{e_n}(B_t) - e_n - L_t^n. \tag{6}$$

The first process is continuous, increasing and nonnegative on every continuous Brownian path. The second is continuous and, by the smooth Itô formula,

$$M_t^n = \int_0^t a_{e_n}(B_u)\, dB_u \quad \text{at each fixed time, almost surely.} \tag{7}$$

The coefficients in this formula are bounded and predictable. The curvature term is integrable on every bounded horizon by (5); thus the formula involves no limiting local time as an input.

Define $\operatorname{sgn}(0) = 0$ and let $M_t = \int_0^t \operatorname{sgn}(B_u)\, dB_u$, initially as an L^2 class for each t. The coefficient is an actual predictable field: it is the bounded pointwise limit of the predictable continuous functions $a_{e_n}(B_u)$. At every x, $a_{e_n}(x) \to \operatorname{sgn}(x)$, and $|a_{e_n}| \le 1$. Dominated convergence on $[0,T] \times \Omega$ and the stochastic isometry give

$$\int_0^T \mathbb{E}|a_{e_n}(B_u) - \operatorname{sgn}(B_u)|^2 du \to 0, \qquad M_T^n \longrightarrow M_T \quad \text{in } L^2. \tag{8}$$

Since $|r_{e_n}(B_T) - e_n - |B_T|| \le e_n$,

$$L_T^n \longrightarrow \ell_T := |B_T| - M_T \quad \text{in } L^2. \tag{9}$$

The closed positive cone in L^2 shows $\ell_T \ge 0$ almost surely. Similarly, $\ell_s \le \ell_t$ as L^2 classes whenever $s \le t$. These fixed-time statements do not yet construct a continuous increasing random path. That distinction motivates the next step.

3 A continuous increasing mass from a summable subsequence

We first give the maximal estimate needed to keep the limiting mass on the original sample space.

Lemma 3.1. *For n, m, $T \ge 0$ and $\varepsilon > 0$,*

$$\varepsilon \mathbb{P}\left(\sup_{u \le T} |M_u^n - M_u^m| > \varepsilon \right) \le \left(\int_0^T \mathbb{E}|a_{e_n}(B_u) - a_{e_m}(B_u)|^2 du \right)^{1/2}. \tag{10}$$

Consequently, L^n is Cauchy in probability uniformly on each compact time interval.

Proof. For a finite ordered set of times including T, $D = M^n - M^m$ is a martingale. Partition the first-crossing event $\max |D| > \varepsilon$ according to its first crossing time u. Each piece belongs to $\mathcal{F}_u$, and $|D_u| \le \mathbb{E}(|D_T| \mid \mathcal{F}_u)$. Summing the integrals over the disjoint pieces gives $\varepsilon \mathbb{P}(\max |D| > \varepsilon) \le \mathbb{E}|D_T|$. An increasing collection of finite sets dense in $[0,T]$ gives the continuous-time estimate by continuity and monotone convergence of events. The stochastic isometry and Cauchy–Schwarz give (10). This argument also proves measurability of the crossing event, using only countably many evaluations.

From (6) and the normalized absolute-value bound,

$$\sup_{u \le T} |L_u^n - L_u^m| \le e_n + e_m + \sup_{u \le T} |M_u^n - M_u^m|. \tag{11}$$

The right side tends to zero in probability by (8) and (10). This proves the claim. □

For each $j \ge 0$, choose N_j such that for $n, m \ge N_j$,

$$\mathbb{P}\left(\sup_{u \le j+1} |L_u^n - L_u^m| > 2^{-j} \right) < 2^{-j}.$$

Set $n_0 = N_0$ and $n_{j+1} = \max\{n_j + 1, N_{j+1}\}$. The consecutive bad events have summable probabilities, so Borel–Cantelli implies that, almost surely, for all sufficiently large j,

$$\sup_{u \le j+1} |L_u^{n_{j+1}} - L_u^{n_j}| \le 2^{-j}. \tag{12}$$

Fix any compact horizon T. For $k > j$ sufficiently large that $j + 1 \ge T$, telescoping bounds the uniform difference by $\sum_{i=j}^{k-1} 2^{-i}$. Completeness of $C([0,T])$ gives a uniform limit. These limits

agree on overlapping compact intervals, giving one continuous path L on $[0,\infty)$. It is increasing, nonnegative and zero at the origin, because every approximant has these properties. The same probability-one event works for all compact horizons, by taking integer horizons first.

The event in (12) is measurable, since uniform crossings of continuous paths are countable evaluation events. Each map $\omega \mapsto L^{n_j}(\omega)$ is measurable into the continuous path space with its compact-convergence topology: evaluations are measurable, and on this separable path space they generate its Borel sigma field. Set the limiting path equal to zero on the exceptional event. The resulting random continuous path is measurable, since it is the path-space limit on a measurable event of convergence. This normalization gives a continuous increasing mass starting at zero on every sample. The source instead centers its Stieltjes cumulative function on exceptional samples; both conventions agree almost surely.

At each fixed T, almost-sure subsequence convergence to L_T and (9) identify

$$L_T = \ell_T \quad \text{almost surely}, \qquad L_T \in L^2. \tag{13}$$

For example, take a further subsequence along which the L^2 convergence is almost sure, and use uniqueness of real limits. Thus (1) holds. Define the continuous process $\widehat{M}_t = |B_t| - L_t$. At each fixed time it is a version of M_t; conditional-expectation identities transfer to this version. It is adapted to $\mathcal{F}$ and a martingale there. The identity

$$|B_t| = \widehat{M}_t + L_t \tag{14}$$

is now a simultaneous pathwise identity, rather than a substitution of a random time into unrelated fixed-time null-set statements.

4 Support of the Stieltjes mass

For a compact interval $[s,t]$ on which $|B_u| \geq \delta > 0$,

$$0 \leq L_t^n - L_s^n = \frac{1}{2}\int_s^t \frac{e_n^2}{(B_u^2 + e_n^2)^{3/2}}\,du \leq \frac{(t-s)e_n^2}{2\delta^3}.$$

Uniform subsequence convergence therefore gives $L_t = L_s$. Every compact interval free of Brownian zeros has a positive such δ, by continuity and compactness. Consequently the Stieltjes measure dL vanishes locally on the open set $\{B \neq 0\}$. Second countability supplies a countable cover by such neighborhoods, so

$$dL(\{s : B_s \neq 0\}) = 0. \tag{15}$$

Continuity of L implies that dL has no atoms, including at zero; its interval masses are $dL((s,t]) = L_t - L_s$, and $dL([0,t]) = L_t$.

Ordinary Brownian time gives no mass to the same zero set. Indeed, $\mathbb{P}(B_s = 0) = 0$ for every $s > 0$, since B_s has a nondegenerate Gaussian law. Tonelli yields

$$\mathbb{E}\int_0^N \mathbf{1}_{\{B_s=0\}}\,ds = \int_0^N \mathbb{P}(B_s = 0)\,ds = 0.$$

The integrand is nonnegative. Taking all integer N gives, on one probability-one event,

$$ds(\{s : B_s = 0\}) = 0. \tag{16}$$

The two different measures in (15) and (16) are exactly what creates physical holding.

Nontriviality follows without a distributional formula for local time. The stochastic integral is centered, so

$$\mathbb{E}L_T = \mathbb{E}|B_T| > 0 \qquad (T > 0). \tag{17}$$

Both terms are integrable by (13). Positivity uses only the nondegenerate Brownian marginal, which makes $|B_T| > 0$ almost surely. We use no assertion that L_T is positive on every sample.

5 Invertible clocks and exact measure transport

We record the deterministic clock mechanism in a form that can be checked before inserting any randomness.

Lemma 5.1. *If $l : [0,\infty) \to [0,\infty)$ is continuous, increasing and $l(0) = 0$, then $A(s) = s + \kappa l(s)$ is a continuous strictly increasing bijection of $[0,\infty)$ onto itself. Its inverse C is continuous, increasing, and satisfies*

$$C(t) \le t, \qquad 0 \le C(t) - C(r) \le t - r \quad (r \le t), \qquad C(t) + \kappa l(C(t)) = t. \tag{18}$$

Proof. For $v > u$, $A(v) - A(u) \ge v - u > 0$. Also $A(0) = 0$ and $A(s) \ge s \to \infty$; the intermediate value theorem gives surjectivity. The inverse of a continuous increasing bijection of an interval is continuous and increasing. Substitute $C(t)$ into A to get the balance identity and $C(t) \le t$. Subtract the balance identities at r, t; the nonnegative mass increment gives the upper increment bound. The lower bound follows from monotonicity. □

Apply this to each mass sample. For fixed κ, t, measurability of the inverse follows from

$$\{C_\kappa(t) < b\} = \{t < A_\kappa(b)\}, \qquad b \ge 0. \tag{19}$$

Each right side is measurable. Continuous time paths and measurable coordinates give jointly measurable clocks and measurable continuous path maps. Evaluation $(p, s) \mapsto p(s)$ is continuous for compact convergence on bounded time neighborhoods. Hence $X_t^\kappa = B_{C_\kappa(t)}$ has measurable coordinates, jointly measurable time/sample evaluation, and a measurable continuous path realization. No random time is selected using an independent copy of B.

On $[0,\infty)$ let ds denote Lebesgue measure with no atom at zero. For any continuous mass path from the construction, define $\eta_\kappa = ds + \kappa\, dL$. Then

$$(A_\kappa)_\# \eta_\kappa = dt. \tag{20}$$

To prove this, test $[0, t]$. Strict invertibility gives $A_\kappa^{-1}([0,t]) = [0, C_\kappa(t)]$, whose η_κ-mass is $C_\kappa(t) + \kappa L_{C_\kappa(t)} = t$. These intervals form a generating pi system and exhaust the half-line by finite-mass integer intervals. Uniqueness of locally finite measures proves (20) on every Borel set. This is a measure identity, so it also applies to each sample's own Borel zero set.

6 Holding time and a strict second-moment budget

Fix a sample on which (15) and (16) hold. For any $\kappa, t \ge 0$, let $Z = \{s : B_s = 0\}$ and $E_t = \{r \le t : X_r^\kappa = 0\}$. These sets are Borel, and

$$A_\kappa^{-1}(E_t) = Z \cap [0, C_\kappa(t)].$$

By (20),

$$\int_0^t \mathbf{1}_{\{X_r^\kappa=0\}} dr = \eta_\kappa(Z \cap [0, C_\kappa(t)]) \\ = \kappa\, dL([0, C_\kappa(t)]) = \kappa L_{C_\kappa(t)} = t - C_\kappa(t).$$

The ordinary-time contribution is zero by (16); concentration of dL on Z gives the second line. The proof uses just two fixed probability-one path properties. Thus it holds simultaneously for all uncountably many κ, t without an uncountable intersection of null sets.

For $\kappa > 0$ and each sample, strict inversion also gives

$$t - C_\kappa(t) > 0 \iff C_\kappa(t) < t \iff t < A_\kappa(t) \iff L_t > 0. \tag{21}$$

For $t > 0$, (17) implies $\mathbb{P}(L_t > 0) > 0$. Since $0 \le H_\kappa(t) \le t$, it is integrable; (21) implies $\mathbb{E}H_\kappa(t) > 0$. The second-moment part of (4) will follow from bounded sampling below. This separates its probabilistic input from the pathwise measure calculation.

7 Bounded sampling with the clock's own Brownian noise

The continuous mass L_s agrees at each fixed time with the raw Brownian-past L^2 class in (1). Hence L_s is $\mathcal{F}_s$-measurable under our ambient null augmentation. The same is true of $A_\kappa(s)$. The inverse is a bounded stopping time because

$$\{C_\kappa(t) \le s\} = \{t \le A_\kappa(s)\} \in \mathcal{F}_s, \qquad C_\kappa(t) \le t. \tag{22}$$

The weak inequality here, as opposed to the strict inequality in (19), is required for the stopping-time sigma field.

We give the bounded sampling argument used in the construction. If Y is a continuous martingale and $\sigma \le \tau \le T$ are stopping times, with integrable compact-time maximum, then

$$\mathbb{E}(Y_\tau \mid \mathcal{F}_\sigma) = Y_\sigma. \tag{23}$$

Round both times up to a dyadic grid, obtaining finite-valued stopping times $\sigma_n \le \tau_n \le T+1$. For a finite grid, (23) follows by writing the sampled difference as a sum of martingale increments multiplied by past-measurable indicators. For any $E \in \mathcal{F}_\sigma$ we also have $E \in \mathcal{F}_{\sigma_n}$, so $\mathbb{E}\mathbf{1}_E Y_{\tau_n} = \mathbb{E}\mathbf{1}_E Y_{\sigma_n}$. Path continuity gives almost-sure convergence of both values. Their absolute values are bounded by $\sup_{u \le T+1} |Y_u|$, so dominated convergence passes this equality to the limit. Stopped values are $\mathcal{F}_\sigma$-measurable: a continuous adapted process is progressively measurable, and its evaluation at a stopping time is measurable in the stopping-time sigma field. This last assertion follows by approximating the continuous path on each deterministic interval by time-step functions, then restricting to $\{\sigma \le s\}$. Integrability and the integral identity for every E prove (23).

The maximum condition holds for the processes used here. The L^2 martingales B and $\widehat{M}$ have integrable compact maxima by the finite-grid L^2 maximal inequality and path continuity. For completeness, that inequality follows by the same first-crossing partition with levels $a > 0$: $a\mathbb{P}(Y^* > a) \le \mathbb{E}(|Y_T|\mathbf{1}_{Y^* > a})$. Integrate twice the resulting inequality in a, then use Cauchy–Schwarz to get $\mathbb{E}(Y^*)^2 \le 2\mathbb{E}(|Y_T|Y^*) \le 2\|Y_T\|_2\|Y^*\|_2$, first with truncated maxima and then by monotone convergence. Thus $\|Y^*\|_2 \le 2\|Y_T\|_2$. Finally,

$$\sup_{u \le T} |B_u^2 - u| \le \sup_{u \le T} B_u^2 + T$$

is integrable. So sampling applies also to the martingale $B_s^2 - s$.

Let $\mathcal{F}_{C_\kappa(t)}$ be the stopping-time sigma field and denote this increasing physical filtration by $\mathcal{H}_t$. Applying (23) at $C_\kappa(r) \le C_\kappa(t)$ gives martingales

$$X_t^\kappa, \qquad (X_t^\kappa)^2 - C_\kappa(t), \qquad \widehat{M}_{C_\kappa(t)} \quad \text{in } \mathcal{H}. \tag{24}$$

Their sampled values are adapted by stopped-value measurability. Sampling the centered square martingale gives $\mathbb{E}(X_t^\kappa)^2 = \mathbb{E}C_\kappa(t)$; its integrability follows from the preceding compact-maximum bound. Combine this equality with (3) to obtain (4), including its strict inequality. The process is centered by the same argument applied to B, so its second moment is indeed its variance.

The pathwise Tanaka identity also becomes

$$|X_t^\kappa| = \widehat{M}_{C_\kappa(t)} + L_{C_\kappa(t)}, \qquad \kappa L_{C_\kappa(t)} = t - C_\kappa(t).$$

This uses (14) on the common path event. Substitution of the random time would not be justified from fixed-time L^2 identities alone.

8 Causal clock recovery and the observed martingales

For a continuous real path p define

$$h(p,t) = \int_0^t \mathbf{1}_{\{p(r)=0\}} dr, \qquad c(p,t) = t - h(p,t). \tag{25}$$

Then $0 \le h(p,t) \le t$ and $0 \le c(p,t) \le t$ for every path. Both functionals are jointly Borel measurable in (p,t). Indeed,

$$(p,t,r) \longmapsto \mathbf{1}_{\{0<r\le t\}} \mathbf{1}_{\{p(r)=0\}}$$

is Borel measurable, since evaluation is continuous and the indicated sets are Borel. Integrating this nonnegative function in r proves measurability by the measurable-integral theorem. The integral is finite, so subtraction in (25) is legitimate. Both functionals depend only on $p|_{[0,t]}$.

To make the raw-past measurability explicit, the map $\omega \mapsto X^\kappa(\omega)|_{[0,t]}$ into $C([0,t])$ is $\mathcal{G}_t^0$-measurable: its evaluations are precisely the coordinates generating $\mathcal{G}_t^0$, and they generate the Borel sigma field of $C([0,t])$. Extend a prefix constantly after t; this is a measurable map back into the full continuous path space. Applying (25) to the extension therefore gives a $\mathcal{G}_t^0$-measurable version of the recovered time. By (3),

$$c(X^\kappa, t) = C_\kappa(t) \quad \text{almost surely.} \tag{26}$$

The actual clock coordinate is ambient measurable. A measurable variable equal almost surely to a raw-past measurable one is measurable in $\mathcal{G}_t^0 \vee \mathcal{N}$. Thus the actual clock is adapted to $\mathcal{G}$, with a causal raw-past version. We do not silently identify these two assertions on unaugmented sigma fields.

For $u \le t$, the coordinate X_u^κ belongs to $\mathcal{H}_u \subseteq \mathcal{H}_t$, so $\mathcal{G}_t^0 \subseteq \mathcal{H}_t$. The elementary shrinking-filtration argument now transfers the first martingale in (24): for $r \le t$,

$$\mathbb{E}(X_t^\kappa \mid \mathcal{G}_r^0) = \mathbb{E}(\mathbb{E}(X_t^\kappa \mid \mathcal{H}_r) \mid \mathcal{G}_r^0) = \mathbb{E}(X_r^\kappa \mid \mathcal{G}_r^0) = X_r^\kappa.$$

It remains a martingale after ambient null augmentation, because adjoining ambient null events changes neither integrals nor conditional expectations up to almost-sure equality.

For the compensated square, use the augmented physical filtration $\mathcal{H}_t \vee \mathcal{N}$. Then $\mathcal{G}_t \subseteq \mathcal{H}_t \vee \mathcal{N}$, and $(X_t^\kappa)^2 - C_\kappa(t)$ is $\mathcal{G}_t$-measurable by (26). The same tower argument transfers the second martingale in (24) to $\mathcal{G}$. This completes every assertion of Theorem 1.1.

9 A balanced coupling of two genuine Brownian coordinates

The same inverse clock has a useful consequence beyond the single sticky path. On the product of two independent Brownian source spaces, let W be independent of the Brownian motion B used to construct L and C_κ. For the rest of this section fix κ and abbreviate

$$C(t) = C_\kappa(t), \qquad D(t) = 2t - C(t), \qquad S_t = B_{C(t)}, \quad V_t = W_{D(t)}.$$

The inverse increment bound in (18) implies that D is continuous and increasing, with $t \le D(t) \le 2t$, $D(0) = 0$, and $C(t) + D(t) = 2t$. Define

$$Y_t^+ = \frac{V_t + S_t}{\sqrt{2}}, \qquad Y_t^- = \frac{V_t - S_t}{\sqrt{2}}. \tag{27}$$

These are measurable continuous paths, by measurable evaluation and the product probability law. The source centers both Brownian paths at their initial values on exceptional samples; they start at zero almost surely, so this convention gives the same path law.

Theorem 9.1. *Each coordinate Y^+ and Y^- has the law of standard Brownian motion, including all finite collections of independent Gaussian increments. For every $t \ge 0$,*

$$\operatorname{Cov}(Y_t^+, Y_t^-) = \mathbb{E}H_\kappa(t) = \mathbb{E}\int_0^t \mathbf{1}_{\{Y_r^+ = Y_r^-\}} dr. \tag{28}$$

For $\kappa, t > 0$ the covariance is strictly positive, so the terminal coordinates are not independent. For every $\kappa, T > 0$ there exists a deterministic $s \in (0, T]$ for which

$$\mathbb{P}(Y_s^+ = Y_s^-) > 0. \tag{29}$$

At such s, the joint law differs from the product of its Gaussian marginals, which gives zero mass to the diagonal.

The assertion that each coordinate is Brownian concerns its own full path law. We do not deduce a Brownian property relative to the joint pair filtration merely from the marginal assertion. Nor do we assert a stochastic flow from this single two-point construction.

Why a single Gaussian marginal does not prove the theorem

Given the entire B sample, the clock $D(t)$ is deterministic and $W_{D(t)}$ is conditionally Gaussian. But $B_{C(t)}$ is sampled at a stopping time depending on B itself. Treating it as an independent Gaussian with variance $C(t)$ is invalid. The compensating Fourier martingale, rather than that invalid conditional independence, gives the cancellation in the proof below.

For $v \in \mathbb{R}$ define the complex process

$$Z_v(s) = \exp\bigl(v^2 s/2 + ivB_s\bigr). \tag{30}$$

This is a continuous complex martingale on every bounded time interval. Indeed, Brownian independent increments give

$$\mathbb{E}(e^{iv(B_t - B_s)} \mid \mathcal{F}_s) = e^{-v^2(t-s)/2},$$

which verifies the martingale identity directly. Its modulus is the deterministic quantity $e^{v^2 s/2}$; its real and imaginary parts therefore satisfy bounded sampling as in Section 7. It is never zero and $|Z_v(s)| \ge 1$.

Lemma 9.2. *For ordered bounded Brownian stopping times $\tau_0 \le \cdots \le \tau_n \le T$ and real $v_0, \dots, v_{n-1}$,*

$$\mathbb{E} \prod_{j=0}^{n-1} \frac{Z_{v_j}(\tau_{j+1})}{Z_{v_j}(\tau_j)} = 1. \tag{31}$$

Proof. Write the product through $j = k-1$ as R_k, with $R_0 = 1$. It is $\mathcal{F}_{\tau_k}$-measurable. Every factor has modulus

$$\exp\bigl(v_j^2(\tau_{j+1} - \tau_j)/2\bigr) \le e^{v_j^2 T/2},$$

so the finite product is bounded by a deterministic constant. Also $1/Z_{v_k}(\tau_k)$ is bounded and $\mathcal{F}_{\tau_k}$-measurable. Bounded sampling of Z_{v_k}, separately in real and imaginary parts, gives

$$\begin{aligned} \mathbb{E}R_{k+1} &= \mathbb{E}\left[\frac{R_k}{Z_{v_k}(\tau_k)} \mathbb{E}(Z_{v_k}(\tau_{k+1}) \mid \mathcal{F}_{\tau_k})\right] \\ &= \mathbb{E}R_k. \end{aligned}$$

All products are integrable by the deterministic bounds. Induction proves (31); independence of the stopping intervals has not been used. □

10 The full finite-increment law by exact compensation

More generally let $a, b \in \mathbb{R}$ satisfy $a^2 = b^2$, and put $Q_t = aV_t + bS_t$. For deterministic ordered times $0 \le t_0 \le \cdots \le t_n$ and real $u_0, \dots, u_{n-1}$, we prove

$$\mathbb{E} \exp\left(i \sum_{j=0}^{n-1} u_j (Q_{t_{j+1}} - Q_{t_j}) \right) = \prod_{j=0}^{n-1} e^{-(u_j a)^2 (t_{j+1} - t_j)}. \tag{32}$$

Condition first on the full B sample. The $D(t_j)$ are then ordered deterministic times for the independent Brownian motion W. Its Gaussian independent increments give the conditional expectation

$$\prod_{j=0}^{n-1} e^{-(u_j a)^2 (D(t_{j+1}) - D(t_j))/2} e^{i u_j b (B_{C(t_{j+1})} - B_{C(t_j)})}. \tag{33}$$

This use of conditioning can equally be written as an iterated integral on the product of the two source spaces. The original integrand has modulus one, so Fubini is valid.

Write $\tau_j = C(t_j)$, $\Delta t_j = t_{j+1} - t_j$ and $\Delta\tau_j = \tau_{j+1} - \tau_j$. Since $\Delta D_j = 2\Delta t_j - \Delta\tau_j$ and $(u_j a)^2 = (u_j b)^2$, each factor in (33) is exactly

$$e^{-(u_j a)^2 \Delta t_j} \exp\Bigl((u_j b)^2 \Delta\tau_j/2 + i u_j b (B_{\tau_{j+1}} - B_{\tau_j}) \Bigr) = e^{-(u_j a)^2 \Delta t_j} \frac{Z_{u_j b}(\tau_{j+1})}{Z_{u_j b}(\tau_j)}. \tag{34}$$

The times τ_j are genuine ordered stopping times, all bounded by t_n. Lemma 9.2 shows that the expectation of the product of the phase ratios is one. Pulling out the deterministic factors proves (32).

For $a = 1/\sqrt{2}$ and $b = \pm 1/\sqrt{2}$, the right side becomes

$$\prod_{j=0}^{n-1} \exp(-u_j^2 \Delta t_j/2),$$

the characteristic function of a vector of independent centered Gaussian increments with variances Δt_j. Equality of characteristic functions determines finite-dimensional probability laws. Thus the increment vectors of Y^+ and Y^- separately have exactly those Brownian laws, including zero-length intervals. Both paths are continuous and start at zero. This proves the full Brownian-coordinate assertion in Theorem 9.1, without inferring it from a one-time Gaussian calculation.

The proof remains valid when $t_0 > 0$: the first phase ratio is conditioned at $C(t_0)$, and the empty product at that time is one. All parameter-dependent bounds are finite for the fixed collection of frequencies and times. No limiting product, random signed law or independence of the Brownian stopping-clock increments is used.

11 Covariance, coincidence and the deterministic-time diagonal

The elementary pathwise identity

$$Y_t^+ - Y_t^- = \sqrt{2} S_t \tag{35}$$

implies

$$\{Y_t^+ = Y_t^-\} = \{S_t = 0\}, \qquad \int_0^t \mathbf{1}_{\{Y_r^+ = Y_r^-\}} dr = H_\kappa(t)$$

on the same common event as (3). The product-space extension preserves that event because the second source is an independent probability factor. Each Brownian coordinate is centered with second moment t. From (35) and the exact single-process variance,

$$\mathbb{E}(Y_t^+ - Y_t^-)^2 = 2\mathbb{E}S_t^2 = 2\mathbb{E}C(t).$$

Expand the square. The product $Y_t^+ Y_t^-$ is integrable by Cauchy–Schwarz, so the calculation is valid and gives

$$2t - 2\mathbb{E}(Y_t^+ Y_t^-) = 2\mathbb{E}C(t), \qquad \mathrm{Cov}(Y_t^+, Y_t^-) = t - \mathbb{E}C(t) = \mathbb{E}H_\kappa(t).$$

This proves (28) and, for $\kappa, t > 0$, strict positivity. Independence would force zero covariance for these L^2 variables, so it is impossible at every such terminal time.

To prove the stronger diagonal statement at some deterministic time in each horizon, Tonelli gives

$$\int_0^T \mathbb{P}(S_s = 0) ds = \mathbb{E}H_\kappa(T) > 0. \tag{36}$$

Joint measurability of S was proved in Section 5. If every deterministic $s \in (0, T]$ had zero boundary probability, the integral would vanish. Hence some such s satisfies $\mathbb{P}(S_s = 0) > 0$, and (35) gives (29). The product of two nondegenerate Gaussian marginals has zero diagonal mass: integrate the singleton mass $\mathbb{P}(Y_s^- = x) = 0$ over the first Gaussian marginal. Thus the joint law cannot be their product. This completes Theorem 9.1.

The covariance conclusion holds at every positive deterministic time, whereas the diagonal conclusion just proved is existential within each positive horizon. The two quantifiers should not be interchanged. Classical sticky Brownian couplings already combine Brownian coordinates with coincidence occupation. Howitt–Warren [3], equations (1)–(2), formulate such couplings using a common-filtration bracket and local time. The present source-mapped claims establish the stated coordinate path laws, covariance and coincidence facts; no identification of a joint bracket or of an entire stochastic flow is silently added.

12 What is established and what remains separate

Clock recovery uses the observed prefix, not a terminal coordinate. It does not construct a strong solution with a prescribed physical-time Brownian driver. Bass's sticky SDE [2] distinguishes weak existence and uniqueness from strong existence; the operational representation (2) does not identify that driver filtration.

The source library separately proves a densely defined closable continuous-function generator relation and a maximum-principle bound. Its range condition and closed-semigroup identification are not supplied by those results. No such missing step is assumed in this paper. In particular, positive two-time conditional operators alone do not establish Chapman–Kolmogorov consistency or an all-start transition family. No stochastic flow, generator-domain classification or uniqueness of the sticky martingale problem is asserted by Theorem 1.1.

13 Formal provenance and reproducibility

All mathematical sources used here are frozen at

`bab688bb6140cd6316c80680095a914ea0d346db`.

The theorem map records exact declarations, complete headers, source hashes and baseline links for the mass, clock, martingale and pair constructions. The original full-build and transitive-axiom receipt and its referenced evidence hashes were checked. No mathematical source was changed and no competing research build was run.

The printed proofs reorganize verified arguments rather than adding formal declarations. Their distinctions between fixed-time L^2 classes, common continuous paths, raw-past versions and augmented actual coordinates are retained. The pair proof supplies the full finite-increment argument beyond Gaussian marginals. The cited primary works provide classical antecedents; no new sticky process, local-time mechanism or priority is claimed.

References

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[3] C. Howitt and J. Warren, *Dynamics for the Brownian web and the erosion flow*, primary preprint (2007). arXiv:math/0702542.